

Java Security Vulnerabilities Analysis - A08_CVE-2023-51656

Here is a snippet of code written in java. I want to check for any potential security vulnerabilities.

```
package org.apache.iotdb.db.sync.receiver.load;

import
org.apache.iotdb.db.concurrent.IoTDBThreadPoolFactory;
import org.apache.iotdb.db.concurrent.ThreadName;
import org.apache.iotdb.db.conf.IoTDBDescriptor;
import
org.apache.iotdb.db.engine.storagegroup.TsFileResource;
import
org.apache.iotdb.db.exception.SyncDeviceOwnerConflictEx
ception;
import org.apache.iotdb.db.sync.conf.SyncConstant;

import org.apache.commons.io.FileUtils;
import org.slf4j.Logger;
import org.slf4j.LoggerFactory;

import java.io.File;
import java.io.FileInputStream;
import java.io.FileOutputStream;
import java.io.IOException;
import java.io.ObjectInputStream;
import java.io.ObjectOutputStream;
import java.util.HashMap;
import java.util.Map;
import java.util.Set;
import java.util.concurrent.ConcurrentHashMap;
import java.util.concurrent.ExecutorService;
import java.util.concurrent.TimeUnit;

/** This class is to manage all FileLoader. */
public class FileLoaderManager {

    private static final Logger LOGGER =
LoggerFactory.getLogger(FileLoaderManager.class);
```

```

private static final int WAIT_TIMEOUT = 2000;

private ConcurrentHashMap<String, IFileLoader>
fileLoaderMap;

private ExecutorService loadTaskRunnerPool;

private Map<String, String> deviceOwnerMap = new
HashMap<>();

private File deviceOwnerFile;

private File deviceOwnerTmpFile;

private FileLoaderManager() {
    String syncSystemDir =
IoTDBDescriptor.getInstance().getConfig().getSyncDir();
    deviceOwnerFile = new File(syncSystemDir,
SyncConstant.DEVICE_OWNER_FILE_NAME);
    deviceOwnerTmpFile = new File(syncSystemDir,
SyncConstant.DEVICE_OWNER_TMP_FILE_NAME);
    try {
        recoverDeviceOwnerMap();
    } catch (IOException | ClassNotFoundException e) {
        LOGGER.error(
            "Can not recover device owner map from file {}",
            new File(syncSystemDir,
SyncConstant.DEVICE_OWNER_FILE_NAME).getAbsolutePath());
    }
}

public static FileLoaderManager getInstance() {
    return FileLoaderManagerHolder.INSTANCE;
}

private void recoverDeviceOwnerMap() throws
IOException, ClassNotFoundException {
    if (deviceOwnerTmpFile.exists()) {
        deviceOwnerFile.delete();
        FileUtils.moveFile(deviceOwnerTmpFile,

```

```

deviceOwnerFile);
    }
    if (deviceOwnerFile.exists()) {
        deSerializeDeviceOwnerMap(deviceOwnerFile);
    }
}

/**
 * Check whether there have conflicts about the device
owner. If there have conflicts, reject the
 * sync process of the sg. Otherwise, update the device
owners and deserialize.
 *
 * @param tsFileResource tsfile resource
 */
public synchronized void
checkAndUpdateDeviceOwner(TsFileResource
tsFileResource)
    throws SyncDeviceOwnerConflictException, IOException
{
    String curOwner =

tsFileResource.getTsFile().getParentFile().getParentFile().get
ParentFile().getName();
    Set<String> deviceSet = tsFileResource.getDevices();
    checkDeviceConflict(curOwner, deviceSet);
    updateDeviceOwner(curOwner, deviceSet);
}

/**
 * Check whether there have conflicts about the device
owner.
 *
 * @param curOwner sender name that want to be owner.
 * @param deviceSet device set
 */
private void checkDeviceConflict(String curOwner,
Set<String> deviceSet)
    throws SyncDeviceOwnerConflictException {
    for (String device : deviceSet) {
        if (deviceOwnerMap.containsKey(device) &&
!deviceOwnerMap.get(device).equals(curOwner)) {

```

```

        throw new SyncDeviceOwnerConflictException(device,
deviceOwnerMap.get(device), curOwner);
    }
}

/**
 * Update the device owners and deserialize.
 *
 * @param curOwner sender name that want to be owner.
 * @param deviceSet device set.
 */
private void updateDeviceOwner(String curOwner,
Set<String> deviceSet) throws IOException {
    boolean modify = false;
    for (String device : deviceSet) {
        if (!deviceOwnerMap.containsKey(device)) {
            deviceOwnerMap.put(device, curOwner);
            modify = true;
        }
    }
    if (modify) {
        serializeDeviceOwnerMap(deviceOwnerTmpFile);
        deviceOwnerFile.delete();
        FileUtils.moveFile(deviceOwnerTmpFile,
deviceOwnerFile);
    }
}

private void deSerializeDeviceOwnerMap(File
deviceOwnerFile)
    throws IOException, ClassNotFoundException {
    try (FileInputStream fis = new
FileInputStream(deviceOwnerFile);
        ObjectInputStream deviceOwnerInput = new
ObjectInputStream(fis)) {
        deviceOwnerMap = (Map<String, String>)
deviceOwnerInput.readObject();
    }
}

private void serializeDeviceOwnerMap(File

```

```

deviceOwnerFile) throws IOException {
    if (!deviceOwnerFile.getParentFile().exists()) {
        deviceOwnerFile.getParentFile().mkdirs();
    }
    if (!deviceOwnerFile.exists()) {
        deviceOwnerFile.createNewFile();
    }
    try (FileOutputStream fos = new
FileOutputStream(deviceOwnerFile, false);
        ObjectOutputStream deviceOwnerOutput = new
ObjectOutputStream(fos)) {
        deviceOwnerOutput.writeObject(deviceOwnerMap);
    }
}

    public void addFileLoader(String senderName, IFileLoader
fileLoader) {
        fileLoaderMap.put(senderName, fileLoader);
    }

    public void removeFileLoader(String senderName) {
        fileLoaderMap.remove(senderName);
    }

    public IFileLoader getFileLoader(String senderName) {
        return fileLoaderMap.get(senderName);
    }

    public boolean containsFileLoader(String senderName) {
        return fileLoaderMap.containsKey(senderName);
    }

    public void addLoadTaskRunner(Runnable taskRunner) {
        loadTaskRunnerPool.submit(taskRunner);
    }

    public void start() {
        if (fileLoaderMap == null) {
            fileLoaderMap = new ConcurrentHashMap<>();
        }
        if (loadTaskRunnerPool == null) {
            loadTaskRunnerPool =

```

```

IoTDBThreadPoolFactory.newCachedThreadPool(ThreadName.LOAD_TSFILE.getName());
    }
}

public void stop() {
    fileLoaderMap = null;
    loadTaskRunnerPool.shutdownNow();
    int totalWaitTime = WAIT_TIMEOUT;
    while (!loadTaskRunnerPool.isTerminated()) {
        try {
            if
(!loadTaskRunnerPool.awaitTermination(WAIT_TIMEOUT,
TimeUnit.MILLISECONDS)) {
                LOGGER.info("File load manager thread pool doesn't
exit after {}ms.", +totalWaitTime);
            }
            totalWaitTime += WAIT_TIMEOUT;
        } catch (InterruptedException e) {
            LOGGER.error("Interrupted while waiting file load
manager thread pool to exit. ", e);
            Thread.currentThread().interrupt();
        }
    }
    loadTaskRunnerPool = null;
}

private static class FileLoaderManagerHolder {

    private static final FileLoaderManager INSTANCE = new
FileLoaderManager();
}
}

```

Below is a security analysis of the provided Java code with potential vulnerabilities and recommendations:

1. Unsafe Deserialization

- **Vulnerability:** The `deSerializeDeviceOwnerMap` method uses `ObjectInputStream` to deserialize the `deviceOwnerMap` from a file. This can lead to **deserialization attacks** if an attacker gains control of the serialized file contents.
- **Risk:** An attacker could craft malicious serialized objects, resulting in remote code execution (RCE).
- **Mitigation:**
 - Validate the deserialized objects with whitelisting or type checks.
 - Use tools like `Gson` or `Jackson` for safer deserialization.
 - Avoid deserialization altogether if possible.

Recommendation:

```
java

if (!(object instanceof HashMap)) {
    throw new IOException("Unexpected object type during deserialization.");
}
```

2. Insecure File Handling

- **Vulnerability:** Temporary files (`deviceOwnerTmpFile`) are used and later renamed, posing a **race condition** risk. An attacker might replace the temp file with malicious content before it is renamed.
- **Risk:** It can allow an attacker to execute arbitrary code or introduce unexpected data.
- **Mitigation:**
 - Use `Files.move()` from `java.nio.file` with the `REPLACE_EXISTING` option for atomic moves.
 - Use **secure temp directories** and proper access controls.

Recommendation:

```
java
```

```
Path tempPath = deviceOwnerTmpFile.toPath();
Path targetPath = deviceOwnerFile.toPath();
Files.move(tempPath, targetPath, StandardCopyOption.ATOMIC_MOVE);
```

3. Insufficient Exception Handling

- **Issue:** The `recoverDeviceOwnerMap()` method logs the error but continues the program even if deserialization fails.
- **Risk:** This might leave the application in an inconsistent state or with missing data.
- **Mitigation:** Ensure the application fails gracefully or handles fallback scenarios properly.

Recommendation:

```
java

catch (IOException | ClassNotFoundException e) {
    LOGGER.error("Failed to recover device owner map.", e);
    throw new RuntimeException("Recovery failed", e); // Consider failing fast
}
```

4. Lack of Synchronization on `fileLoaderMap`

- **Issue:** `ConcurrentHashMap` is thread-safe for individual operations, but compound operations like `containsFileLoader` and `addFileLoader` are not synchronized.
- **Risk:** Inconsistencies can arise if multiple threads modify the map simultaneously.
- **Mitigation:** Synchronize compound operations or use higher-level abstractions such as `computeIfAbsent`.

Recommendation:

```
java
```

```
public synchronized void addFileLoader(String senderName, IFileLoader fileLoader) {
    fileLoaderMap.put(senderName, fileLoader);
}
```

5. Hardcoded Wait Time for Thread Pool Shutdown

- **Issue:** A hardcoded wait time may cause a **Denial of Service (DoS)** if the thread pool does not terminate as expected.
- **Mitigation:** Consider configuring the wait time dynamically or using an exponential backoff strategy to prevent blocking.

6. Inadequate Resource Cleanup

- **Issue:** If `FileOutputStream` or `FileInputStream` fail mid-operation, the streams are not properly closed, which can lead to **resource leaks**.
- **Mitigation:** Use **try-with-resources** to ensure proper cleanup of resources.

Recommendation:

```
java

try (FileOutputStream fos = new FileOutputStream(deviceOwnerFile);
    ObjectOutputStream oos = new ObjectOutputStream(fos)) {
    oos.writeObject(deviceOwnerMap);
}
```

7. Lack of Input Validation for Device Names

- **Issue:** No validation is performed on device owner names or device sets. This could lead to **injection attacks** or invalid data.

- **Mitigation:** Implement strict input validation to sanitize or limit the input size and content.

Summary of Potential Vulnerabilities and Recommendations

Vulnerability	Risk	Recommendation
Unsafe deserialization	RCE via malicious input	Use safe deserialization methods or checks
Insecure file handling	Race conditions / data corruption	Use atomic file operations
Insufficient exception handling	Program in inconsistent state	Fail fast or use fallback mechanisms
Lack of synchronization	Data inconsistency	Synchronize or use <code>computeIfAbsent</code>
Hardcoded wait time	DoS due to blocked threads	Use dynamic or backoff strategies
Inadequate resource cleanup	Resource leaks	Use try-with-resources
No input validation	Injection attacks / invalid data	Implement input validation

Addressing these issues will enhance the security and reliability of the code.

Provide the exact part of the code along with line numbers that are vulnerable or have security issues.

Below is the **line-by-line security review** of your code with the exact **locations of vulnerabilities** or potential security issues:

1. Unsafe Deserialization

Issue: Unsafe use of `ObjectInputStream` for deserialization without validating the object type.

- **File:** `FileLoaderManager.java`
- **Lines:** 119-123

java

```
119: try (FileInputStream fis = new FileInputStream(deviceOwnerFile);
120:      ObjectInputStream deviceOwnerInput = new ObjectInputStream(fis)) {
121:     deviceOwnerMap = (Map<String, String>) deviceOwnerInput.readObject(); //
Unsafe cast
122: }
123: }
```

Fix: Add validation to ensure that only valid objects are deserialized.

java

```
Object obj = deviceOwnerInput.readObject();
if (!(obj instanceof HashMap)) {
    throw new IOException("Unexpected object type during deserialization.");
}
```

2. Insecure File Handling (Race Condition)

Issue: `FileUtils.moveFile()` creates a **race condition** when moving the temp file to the target.

- **File:** `FileLoaderManager.java`
- **Lines:** 62-63 and 105

java

```
62: FileUtils.moveFile(deviceOwnerTmpFile, deviceOwnerFile); // Non-atomic move
...
105: FileUtils.moveFile(deviceOwnerTmpFile, deviceOwnerFile); // Repeated risk
```

Fix: Use atomic move operations from `java.nio.file`.

```
java

Files.move(deviceOwnerTmpFile.toPath(), deviceOwnerFile.toPath(),
StandardCopyOption.ATOMIC_MOVE);
```

3. Insufficient Exception Handling during Recovery

Issue: Errors in recovery are logged but not rethrown, potentially leaving the program in an inconsistent state.

- **File:** `FileLoaderManager.java`
- **Lines:** 45-51

```
java

45: try {
46:     recoverDeviceOwnerMap();
47: } catch (IOException | ClassNotFoundException e) {
48:     LOGGER.error(
49:         "Can not recover device owner map from file {}",
50:         new File(syncSystemDir,
SyncConstant.DEVICE_OWNER_FILE_NAME).getAbsolutePath());
51: }
```

Fix: Fail fast or apply a fallback strategy.

```
java

} catch (IOException | ClassNotFoundException e) {
    LOGGER.error("Failed to recover device owner map.", e);
    throw new RuntimeException("Recovery failed", e); // Fail fast
}
```

4. Lack of Synchronization on `fileLoaderMap` Operations

Issue: Compound operations on `ConcurrentHashMap` are not synchronized, leading to race conditions.

- **File:** `FileLoaderManager.java`
- **Lines:** 138-141 and 144-147

java

```
138: public void addFileLoader(String senderName, IFileLoader fileLoader) {
139:     fileLoaderMap.put(senderName, fileLoader); // Not synchronized
140: }
...
144: public IFileLoader getFileLoader(String senderName) {
145:     return fileLoaderMap.get(senderName); // Unsynchronized access
146: }
```

Fix: Synchronize methods where multiple operations on `fileLoaderMap` occur.

java

```
public synchronized void addFileLoader(String senderName, IFileLoader fileLoader) {
    fileLoaderMap.put(senderName, fileLoader);
}

public synchronized IFileLoader getFileLoader(String senderName) {
    return fileLoaderMap.get(senderName);
}
```

5. Hardcoded Wait Time for Thread Pool Shutdown

Issue: Fixed wait time in `stop()` method may block indefinitely, resulting in potential DoS.

- **File:** `FileLoaderManager.java`
- **Lines:** 167-177

java

```

167: int totalWaitTime = WAIT_TIMEOUT;
168: while (!loadTaskRunnerPool.isTerminated()) {
169:     try {
170:         if (!loadTaskRunnerPool.awaitTermination(WAIT_TIMEOUT,
TimeUnit.MILLISECONDS)) {
171:             LOGGER.info("File load manager thread pool doesn't exit after {}ms.",
+totalWaitTime);
172:         }
173:         totalWaitTime += WAIT_TIMEOUT;
174:     } catch (InterruptedException e) {
175:         LOGGER.error("Interrupted while waiting file load manager thread pool to
exit. ", e);
176:         Thread.currentThread().interrupt();
177:     }

```

Fix: Consider using an **exponential backoff strategy** or limit the maximum wait time.

6. Inadequate Resource Cleanup

Issue: Streams in `serializeDeviceOwnerMap` and `deSerializeDeviceOwnerMap` are not properly closed in case of an exception.

- **File:** `FileLoaderManager.java`
- **Lines:** 119-123 and 126-130

```

java

119: try (FileInputStream fis = new FileInputStream(deviceOwnerFile);
120:     ObjectInputStream deviceOwnerInput = new ObjectInputStream(fis)) {
...
126: try (FileOutputStream fos = new FileOutputStream(deviceOwnerFile, false);
127:     ObjectOutputStream deviceOwnerOutput = new ObjectOutputStream(fos)) {
128:     deviceOwnerOutput.writeObject(deviceOwnerMap);
129: }

```

Fix: These lines already use **try-with-resources**, which is the correct approach. However, ensure this pattern is consistently applied throughout the code.

7. Lack of Input Validation for Device Names

Issue: No validation is performed on the `curOwner` or `deviceSet` inputs. This could result in injection attacks or processing of invalid data.

- **File:** `FileLoaderManager.java`
- **Lines:** 85-93

```
java

85: String curOwner =
tsFileResource.getTsFile().getParentFile().getParentFile().getParentFile().getName();
86: Set<String> deviceSet = tsFileResource.getDevices();
87: checkDeviceConflict(curOwner, deviceSet); // No validation on curOwner or
deviceSet
88: updateDeviceOwner(curOwner, deviceSet);
```

Fix: Implement input validation for the `curOwner` and `deviceSet`.

```
java

if (curOwner == null || curOwner.isEmpty() || !curOwner.matches("[a-zA-Z0-9_-]+")) {
    throw new IllegalArgumentException("Invalid owner name: " + curOwner);
}
```

Summary of Vulnerabilities with Exact Code Locations

Vulnerability	File	Line Numbers	Issue
Unsafe deserialization	<code>FileLoaderManager</code>	119-123	RCE risk with unvalidated deserialization
Insecure file handling (race)	<code>FileLoaderManager</code>	62, 105	Race condition with <code>FileUtils.moveFile()</code>
Insufficient exception handling	<code>FileLoaderManager</code>	45-51	Recovery failure not handled properly

Vulnerability	File	Line Numbers	Issue
Lack of synchronization	FileLoaderManager	138-141, 144-147	Race conditions in <code>fileLoaderMap</code> operations
Hardcoded wait time	FileLoaderManager	167-177	DoS risk due to fixed wait time
Resource cleanup issues	FileLoaderManager	119-123, 126-130	Streams correctly handled, ensure consistency
No input validation	FileLoaderManager	85-93	Injection risk with unchecked inputs

Addressing these specific vulnerabilities will make your code more secure and robust.

Please analyze the code for common security vulnerabilities if present and map them to OWASP Top 10 Web Application Security Risk category/categories.

Here's a detailed analysis of the code snippet you provided, mapped to the relevant **OWASP Top 10 Web Application Security Risks**. This will help you understand the vulnerabilities better and how they align with the OWASP guidelines.

1. Unsafe Deserialization

Location:

- File: `FileLoaderManager.java`
- Lines: 119-123

Description:

The use of `ObjectInputStream` to deserialize data from files without proper validation opens the application to **remote code execution (RCE)** attacks if malicious input is processed.

OWASP Top 10 Mapping:

- **A08:2021 – Software and Data Integrity Failures**

Deserialization without proper validation can allow attackers to inject malicious objects.

Mitigation:

- Validate the type of the object before casting.
 - Avoid using Java serialization unless strictly necessary, and prefer safer alternatives like JSON or YAML.
-

2. Insecure File Handling (Race Condition)

Location:

- **File:** `FileLoaderManager.java`
- **Lines:** 62-63, 105

Description:

The use of `FileUtils.moveFile()` creates a **race condition**. If an attacker can replace the temp file (`deviceOwnerTmpFile`) between creation and renaming, it can lead to **malicious data injection or corruption**.

OWASP Top 10 Mapping:

- **A05:2021 – Security Misconfiguration**
Using non-atomic file operations can result in unexpected system behavior, leading to data integrity failures.

Mitigation:

- Use `java.nio.file.Files.move()` with `StandardCopyOption.ATOMIC_MOVE` to perform atomic file moves.
 - Use directories with strict access control to store sensitive or temporary files.
-

3. Insufficient Exception Handling

Location:

- File: `FileLoaderManager.java`
- Lines: 45-51

Description:

When `recoverDeviceOwnerMap()` fails, the exception is only logged, and the program continues to run. This can leave the system in an inconsistent state or expose it to **faulty** behavior.

OWASP Top 10 Mapping:

- **A01:2021 – Broken Access Control**
Inconsistent system state can lead to privilege escalation or incorrect permission handling.
- **A10:2021 – Server-Side Request Forgery (SSRF)** (potential)
If incorrect recovery exposes internal systems, it can create SSRF or privilege issues.

Mitigation:

- Fail fast by re-throwing exceptions or applying fallback mechanisms.
 - Ensure logging is consistent and does not leak sensitive information.
-

4. Lack of Synchronization on `ConcurrentHashMap` Operations

Location:

- File: `FileLoaderManager.java`
- Lines: 138-141, 144-147

Description:

While `ConcurrentHashMap` provides thread-safety for individual operations, **compound operations** such as `containsFileLoader` and `addFileLoader` are not synchronized. This can result in **race conditions** and inconsistent states.

OWASP Top 10 Mapping:

- **A04:2021 – Insecure Design**

Improper handling of concurrency can lead to incorrect business logic or privilege escalation.

Mitigation:

- Use synchronized methods or higher-level abstractions such as `computeIfAbsent` to ensure thread-safety.
-

5. Hardcoded Wait Time for Thread Pool Shutdown (Potential DoS)

Location:

- **File:** `FileLoaderManager.java`
- **Lines:** 167-177

Description:

The thread pool shutdown uses a **hardcoded wait time**. If the threads do not exit properly, the system could be blocked indefinitely, leading to a **Denial of Service (DoS)**.

OWASP Top 10 Mapping:

- **A06:2021 – Vulnerable and Outdated Components**

Relying on blocking operations without graceful handling can degrade performance and availability.

Mitigation:

- Use an **exponential backoff strategy** for waiting.
 - Set a reasonable upper limit on wait times.
-

6. Inadequate Resource Cleanup

Location:

- File: `FileLoaderManager.java`
- Lines: 119-123, 126-130

Description:

Although `try-with-resources` is used correctly, you must ensure that **all input/output operations** throughout the code follow this pattern to avoid **resource leaks**. Leaked resources (e.g., file handles or streams) can be **exploited in DoS attacks**.

OWASP Top 10 Mapping:

- **A06:2021 – Vulnerable and Outdated Components**
Resource mismanagement can degrade performance and cause availability issues.

Mitigation:

- Ensure that **all I/O operations** throughout the code use `try-with-resources`.
 - Monitor and limit the number of open files/streams.
-

7. Lack of Input Validation for Device Names

Location:

- File: `FileLoaderManager.java`
- Lines: 85-93

Description:

The `curOwner` and `deviceSet` inputs are not validated. This can result in **injection attacks** or the processing of **malicious input**, potentially leading to privilege escalation or business logic vulnerabilities.

OWASP Top 10 Mapping:

- **A03:2021 – Injection**
Lack of input validation could lead to injection attacks (e.g., file path injection, SQL injection).

- **A04:2021 – Insecure Design**

Weak input validation exposes the system to undefined behavior or logic flaws.

Mitigation:

- Use strict validation on `curOwner` and `deviceSet` to ensure only valid characters are accepted.
- Sanitize the input to prevent path traversal or injection attacks.

Summary of Vulnerabilities and OWASP Top 10 Mapping

Vulnerability	Location	OWASP Top 10 Risk	Line Numbers
Unsafe Deserialization	<code>deSerializeDeviceOwnerMap()</code>	A08:2021 – Software and Data Integrity Failures	119-123
Insecure File Handling (Race)	<code>recoverDeviceOwnerMap()</code>	A05:2021 – Security Misconfiguration	62-63, 105
Insufficient Exception Handling	<code>recoverDeviceOwnerMap()</code>	A01:2021 – Broken Access Control	45-51
Lack of Synchronization (Race)	<code>fileLoaderMap</code> Operations	A04:2021 – Insecure Design	138-141, 144-147
Hardcoded Wait Time (Potential DoS)	<code>stop()</code>	A06:2021 – Vulnerable and Outdated Components	167-177
Inadequate Resource Cleanup	Serialization Methods	A06:2021 – Vulnerable and Outdated Components	119-123, 126-130
Lack of Input Validation	<code>checkAndUpdateDeviceOwner()</code>	A03:2021 – Injection	85-93

Conclusion

The code contains several vulnerabilities, including **unsafe deserialization**, **race conditions**, and **insufficient input validation**, which align with multiple OWASP Top 10 risks. Addressing these issues will significantly improve the security of the application. Specifically:

1. **Mitigate deserialization risks** by validating object types or using safer formats.
2. **Use atomic file operations** to prevent race conditions.
3. **Implement input validation** to avoid injection attacks.
4. **Synchronize compound operations** on shared data structures.
5. **Handle exceptions** gracefully to avoid inconsistent states.

By following these recommendations, you can mitigate critical security risks and align the code with secure coding practices.